

Designing and Evaluating Gen-AI for Cultural Resilience

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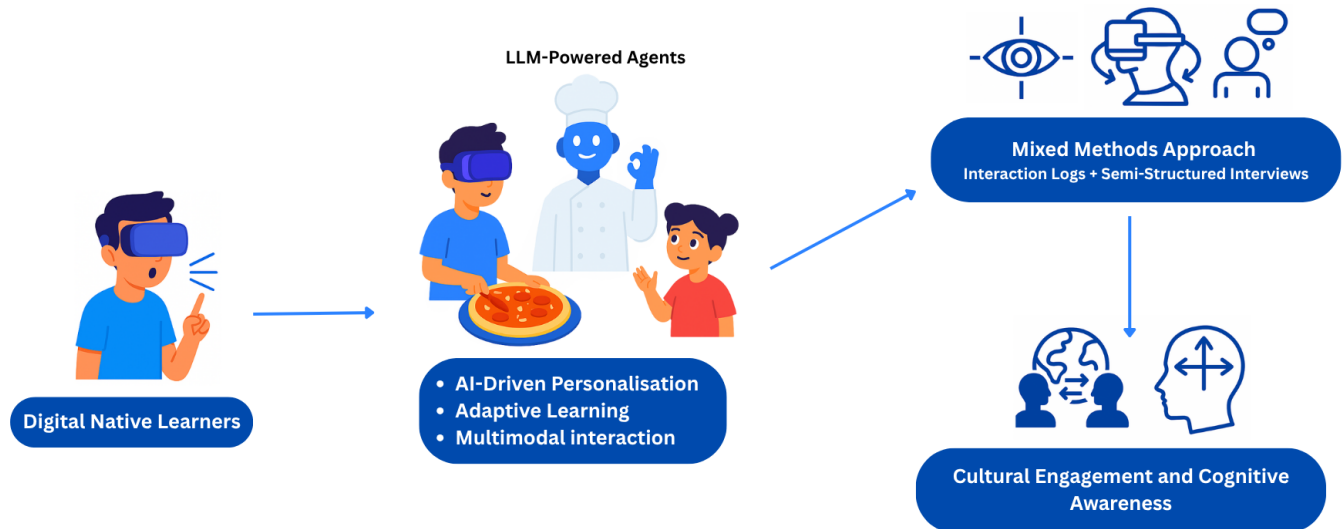


Figure 1: An overview of my research pipeline. Digital-native users interact with Gen-AI-powered agents embedded in virtual cultural learning systems. These agents support narrative personalisation, adaptive guidance, and multimodal interaction (e.g., voice-based dialogue via speech-to-text (STT)/text-to-speech (TTS)). Using a mixed-methods evaluation (interaction logs and semi-structured interviews), I investigate how these systems affect cultural engagement, identity reflection, and cognitive awareness, ultimately informing design considerations for culturally responsive and resilient Gen-AI systems.

Abstract

As cultural traditions face disruption from generational gaps, climate change, and political conflicts, there is a growing need for new ways to engage younger learners in cultural preservation. Generative AI (Gen-AI) and immersive technologies offer promising tools, but little is known about how to design them effectively for cultural learning. My research explores how Gen-AI can support cultural revitalisation through interactive, emotionally resonant systems. I design and evaluate immersive learning experiences, such as voice-based storytelling and guided procedural tasks, using Gen-AI agents to personalise narratives and adapt to user input. Through user studies, I examine how these systems affect engagement, cultural reflection, and cognitive load. Looking ahead, my dissertation investigates how Gen-AI influences identity, bias, and

social perception in cultural contexts. This work contributes design insights for building more inclusive, reflective, and culturally responsive AI systems.

CCS Concepts

• **Applied computing** → **Arts and humanities**; • **Human-centered computing** → **User studies**; *Empirical studies in interaction design*.

Keywords

intangible culture heritage, virtual reality, generative artificial intelligence

ACM Reference Format:

Ka Hei Carrie Lau. 2025. Designing and Evaluating Gen-AI for Cultural Resilience. In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25)*, October 13–17, 2025, Canberra, ACT, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3716553.3750819>

1 Introduction

Generative artificial intelligence (Gen-AI) systems are increasingly embedded in daily life, reshaping how people access information, form social bonds, and engage with culture. While these technologies open up new possibilities for learning and creativity, they also

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ICMI '25, Canberra, ACT, Australia

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ACM ISBN 979-8-4007-1499-3/25/10

<https://doi.org/10.1145/3716553.3750819>

raise important questions about the preservation of intangible cultural heritage (ICH) and the broader social and cognitive effects of AI-mediated interaction. According to UNESCO¹, ICH refers to “the practices, representations, expressions, knowledge, and skills that communities, groups and, in some cases, individuals recognise as part of their cultural heritage.” UNESCO further highlights that ICH must be continuously recreated and transmitted to remain relevant, and that without appropriate safeguarding, many forms of ICH risk disappearing or being forgotten².

Traditional methods of cultural transmission are often static, text-based, and didactic, struggling to engage younger, digitally native audiences. This gap threatens cultural continuity, especially in communities vulnerable to generational disconnect, environmental disruption, or sociopolitical instability. In response, my early research explored how Gen-AI-powered virtual reality (VR) systems might support cultural preservation through interactive, emotionally resonant learning experiences. While the use of Gen-AI and VR in cultural heritage remains an emerging research area, a few promising examples have begun to demonstrate its potential [5, 7]. My work builds on and extends these efforts by investigating how adaptive and multimodal Gen-AI systems can enhance engagement, cultural reflection, and identity awareness.

In this dissertation, I conceptualise **resilience** in two ways: (1) *cultural resilience*, referring to the continuity and revitalisation of ICH through emerging technologies, and (2) *cognitive and social resilience*, which involves users’ ability to reflect critically on AI systems, maintain cultural identity, and navigate AI-mediated experiences with awareness and agency.

Building on this foundation, my research expands from the question: “What can Gen-AI do for cultural education?” to “How does Gen-AI shape the way we think, relate, and construct meaning?” As Gen-AI agents become more conversational and personalised, I explore how interactions with these systems influence identity, reflection, and social perception. For example, do AI agents reinforce existing cultural or social biases? How might AI-mediated storytelling affect a user’s sense of belonging?

Rather than proposing fixed technical solutions, my work aims to surface key design considerations and critical reflections. I examine how Gen-AI systems can be designed to promote awareness of cultural and cognitive resilience, supporting more inclusive, ethically grounded approaches to AI in educational and cultural domains. To guide this research dissertation, I investigate the following research questions (RQs):

- (1) How can Gen-AI be designed and implemented to effectively revitalise and teach intangible cultural heritage?
- (2) Which mediums and modalities are most effective for deploying Gen-AI to enhance user cultural engagement and learning outcomes?
- (3) What are the cognitive and social impacts of interacting with Gen-AI agents across different platforms, and how can insights from cognitive science inform their design and evaluation?

¹<https://ich.unesco.org/en/convention#art2>, last accessed May 27, 2025

²<https://ich.unesco.org/en/safeguarding-00012>, last accessed June 11, 2025

2 Related Work

My research intersects three domains: the digital revitalisation of ICH, adaptive learning and personalisation using AI, and the social and cognitive effects of Human-AI interaction as shown in Figure 2. Each of these areas has made important contributions, but key gaps remain in how they are connected, particularly in evaluating cultural and cognitive impacts in real-time, interactive settings.

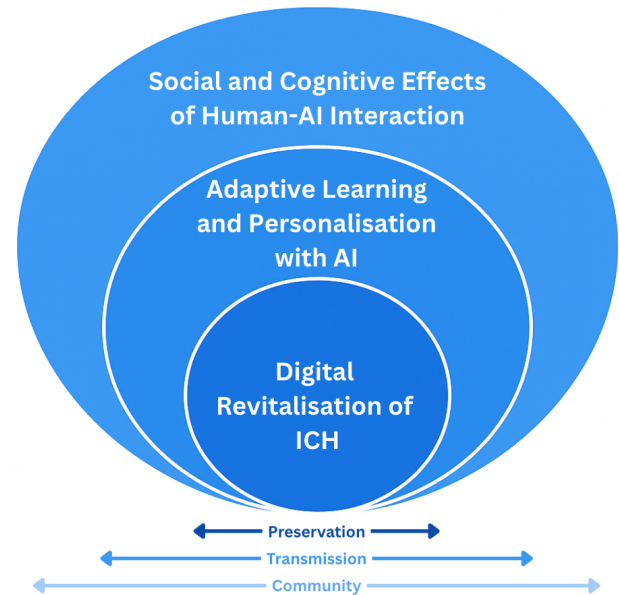


Figure 2: The layered structure of my research. Each layer builds upon the previous: starting from immersive systems for cultural revitalisation, adding adaptive AI personalisation, and extending to evaluate cognitive and social dynamics in Human-AI interaction.

2.1 Digital Revitalisation of Intangible Cultural Heritage

A growing body of research shows that digital technologies, such as virtual museums, extended reality (XR), and immersive reconstructions, not only help preserve ICH but also enhance visitor engagement [2, 8]. Projects like Bamboo-Weaving VR and the Virtual Dance Museum demonstrate how embodied interaction can make traditional practices more accessible and meaningful [1, 18]. However, many current systems remain relatively static and non-interactive, potentially limiting their appeal to younger, tech-savvy users [13]. While VR enables more dynamic and embodied cultural experiences, few studies examine how Gen-AI can shape cultural understanding or support adaptive, real-time narratives. Gen-AI offers the ability to produce culturally resonant content on demand, allowing for more guided, responsive, and interactive learning experiences in virtual environments. This research addresses that gap by integrating Gen-AI into VR to support culturally responsive storytelling and procedural learning, where users follow step-by-step cultural practices in contextually meaningful ways.

2.2 Personalisation and Adaptive Learning with Gen-AI

As digital heritage systems evolve, there is growing interest in personalising user experiences to better match individual preferences, cultural backgrounds, and learning needs. Personalisation has been shown to improve learning engagement and motivation, especially in educational settings [3]. In cultural contexts, adaptive systems can tailor narratives or experiences to a visitor's prior knowledge or interests, enhancing relevance and reducing cognitive overload [6]. Recent work explores how Gen-AI-driven systems can dynamically generate or curate educational content based on user profiles or behaviour, enabling more personalised and context-aware experiences [12]. However, few studies examine how these systems can support reflective or procedural learning in the context of ICH, particularly in VR. This research extends prior work on adaptive AI by exploring how it can also be used to guide culturally aware and adaptive learning experiences in immersive environments.

2.3 Social and Cognitive Implications of Gen-AI Interaction

Recent research in Human-Computer Interaction (HCI) and cognitive science shows that users engage with AI agents in socially meaningful ways, including trust, anthropomorphism, and in-group bias [14, 15]. Studies on large language models (LLMs) have also raised concerns about bias and cultural misrepresentation [16]. However, most work focuses on prescribed chatbots, with limited attention to cultural or educational settings. Few studies consider how users' cultural identity or background influences these interactions. There is limited research on how real-time, speech-based Gen-AI agents affect users' cognitive framing, social perception, and cultural identification. This work addresses that gap by examining how identity cues and cultural biases emerge in conversation with Gen-AI agents, which is critical for understanding whether behaviours present in human-human interaction carry over to human-AI interaction.

3 Research Approach and Methods

This dissertation employs a layered, mixed-methods approach to explore how Gen-AI can support cultural resilience by: (1) designing immersive systems for revitalising and transmitting ICH; (2) evaluating adaptive personalisation effects on user engagement and cultural connection; and (3) investigating cognitive and social impacts of human-Gen-AI interactions, particularly regarding cultural identity and biases. Quantitative methods (e.g., eye-tracking, questionnaires, interaction logs) allow reproducible and comparable insights on usability and learning effectiveness, while qualitative methods (e.g., interviews, thematic analysis) capture nuanced user experiences, cultural reflections, and identity considerations.

3.1 Phase 1: Designing Immersive Systems for Cultural Revitalisation

Addressing RQ1, this phase explored how immersive Gen-AI-powered systems can revitalise ICH through two core design dimensions: (1) narrative adaptation (in *Anansi VR*) and (2) procedural scaffolding (in *Neapolitan Pizza VR*). We hypothesised that personalised

narrative adaptation would enhance cultural engagement and reflection, while adaptive scaffolding would improve learning focus and reduce cognitive load during procedural cultural tasks.

- **Anansi the Spider VR** [11], a Gen-AI-powered narrative experience for exploring oral storytelling traditions.
- **Neapolitan Pizza VR** [10], a procedural learning simulation of cultural cuisine, guided by an adaptive AI assistant.

Both systems were built in Unity and integrated with GPT-4 via voice interaction. Both studies adopted a mixed-method approach, quantitative metrics (e.g., eye-tracking, interaction logs, task performance) with qualitative feedback (e.g., semi-structured interviews) to evaluate engagement, learning, and cultural reflection.

3.2 Phase 2: Exploring Adaptive Personalisation and Modalities

To investigate RQ2, this phase evaluates how different Gen-AI modalities (speech-based vs. rule-based agents) and levels of adaptive feedback affect user engagement and cognitive load. We hypothesised that Gen-AI-based agents and moderately adaptive feedback would lead to higher engagement and better task performance, while excessive adaptivity might reduce exploratory behaviour.

- Experimental comparisons between rule-based and Gen-AI-based agents in a curling-themed VR environment for Scottish cultural practice.
- Analysis of adaptive feedback levels in *Neapolitan Pizza VR* to study how personalisation impacts cognitive load and exploration.

Findings from these studies inform how Gen-AI systems can be designed to balance guidance with user autonomy, supporting more relevant and adaptive cultural learning.

Phase 3: Evaluating Cognitive and Social Impacts of Gen-AI

To answer RQ3, the third research layer focuses on how Gen-AI influences users' cognitive framing, social perception, and cultural identification. We hypothesise that user identity (e.g., cultural background) and perceived group alignment of the AI agent will influence trust, perceived fairness, and cultural bias.

- Developing a real-time, speech-based Gen-AI agent to study phenomena such as in-group favouritism, identity projection, and cultural bias.
- Combining behavioural analysis (e.g., visual attention) to examine user perception.

This layer provides insights into the ethical, social, and cognitive dimensions of Gen-AI systems in cultural and educational settings.

4 Preliminary Results

My dissertation research has produced three user studies examining how Gen-AI can support the revitalisation of ICH through immersive and adaptive systems. These studies address gaps identified in prior literature, particularly around cultural engagement, adaptive guidance, and cultural reflection in AI-mediated experiences.

4.1 Personalisation Enhances Cultural Engagement

In the *Anansi the Spider VR* study ($n = 48$), participants experienced either a personalised or non-personalised Gen-AI-driven oral storytelling session, presented in either VR or desktop [11]. Results showed that Gen-AI personalisation significantly enhanced engagement and interest in cultural learning. Participants in the personalised condition reported significantly higher engagement scores. Qualitative feedback further revealed that addressing users by name and enabling narrative agency deepened emotional resonance and cultural reflection. These findings suggest that narrative adaptation and personalisation are valuable design levers for enhancing cultural engagement.

4.2 Moderate AI-Adaptivity Supports Focused Learning

In the *Neapolitan Pizza VR* study ($n = 54$), we explored how different levels of AI adaptivity affect learners' attention and physical engagement during a procedural cooking task. Eye-tracking analysis showed a significant difference in dwell time on the avatar across adaptivity conditions, $H(2) = 10.11$, $p = .006$. Participants in the Moderate adaptivity condition spent more time attending to the avatar (Median = 14.47%, IQR = 9.04–20.84) compared to No adaptivity (Median = 7.20%, IQR = 4.71–9.68), with Dunn's post-hoc test confirming this difference ($p = .005$). Head movement analysis further revealed that Moderate adaptivity reduced excessive exploratory motion, $H(2) = 7.74$, $p = .021$. Median movement speed was lower in Moderate adaptivity (0.329 m/s, IQR = 0.270–0.354) compared to No adaptivity (0.353 m/s, IQR = 0.349–0.384), $p = .029$.

These findings suggest that moderate adaptivity fosters greater attentional focus and more efficient exploratory behavior during procedural cultural learning.

4.3 Conversational Modality Affects User Experience

In the *Scottish Bonspiel VR* study ($n = 36$), participants compared a predefined rule-based curling coach and a Gen-AI (ChatGPT-based) coach [9]. Overall user experience was rated significantly higher for the Gen-AI agent ($M = 5.05$, $SD = 1.68$) compared to the predefined chatbot ($M = 3.24$, $SD = 1.35$), $U = 63.5$, $p = .002$, $r = 0.518$.

4.4 Cultural Identity Shapes Interpretation

Across these studies, early evidence suggests that user identity, such as cultural background and familiarity with the content, influences how Gen-AI responses are interpreted and trusted. This motivates my current work on investigating cognitive and social dynamics in Gen-AI interaction, particularly regarding cultural bias and identity salience.

5 Contributions

My dissertation explores how Gen-AI can be designed and evaluated to support cultural heritage learning through immersive, multimodal systems. Most of the research has been completed, with results published at venues such as ACM ICMI, ECCV (Workshop) and the Augmented Humans (AHs) International Conference.

While the studies address different cultural contexts and technical approaches, a central theme emerges around how Gen-AI can be used to support engagement, adaptation, and reflection in cultural learning. Through this Doctoral Consortium, I seek feedback on how to refine and position this narrative within the broader ICMI community. Below, I outline my contributions using the HCI contribution framework [17].

Artifact Contributions My research includes proposing a system architecture to integrate Unity-based virtual environments with LLMs (e.g., GPT-4) for real-time cultural storytelling and procedural instruction. These systems include STT and TTS integration, enabling natural voice-based interaction with Gen-AI agents in VR [4]. This architecture offers a responsive, multimodal platform for evaluating the potential of voice-interactive AI in cultural education.

Empirical Contributions I conducted three user studies across all systems to investigate how personalisation, modality, and adaptivity influence user experience. Quantitative and qualitative findings highlight effects on engagement, cognitive load, and cultural reflection, providing evidence for how system design can impact learning engagement and identity-related outcomes.

Design Contributions My ongoing work explores how identity, bias, and group affiliation shape users' interactions with Gen-AI agents in real time. Early findings suggest that user background can influence trust and perceived fairness, especially in culturally embedded scenarios. These insights contribute to broader conversations about inclusive AI design and the emergence of culturally sensitive phenomena in human–AI interaction.

Looking ahead, I aim to translate these findings into design guidelines for Gen-AI systems in educational and heritage contexts. My work may also support cultural practitioners in leveraging Gen-AI for education and the transmission of traditional knowledge.

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